

The Effects of Climate Change (normal values/anomalous values) on Fire Duration, in Greece Areas.

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Abstract

In this study, we examine and highlight the effects of climate change on forest fires duration that have afflicted Greece over the past 25 years (2000–2025). Thus, we selected three distinct regions of Greece, spanning a distance of approximately 1,000 kilometers, in order to ensure impartial findings that are not confined to a single area and, consequently to avoid creating a misleading impression. The first area of interest is Attica, with 2,100 rows of climatological data and fire duration records. The second area lies further south and is Crete, with 2,893 rows of climatological data and fire duration records. The third area of interest is located further north and is Eastern Macedonia & Thrace, with 2,919 rows of climatological data and fire duration records. The baseline for our analysis was established using climatological data from 1980–2010 for the regions of Attica and Crete, and from 1992–2022 for Eastern Macedonia and Thrace, reflecting the earliest available records for that area. These datasets were processed to derive monthly mean values representing the climatological normal's. We then compared these normal's with the meteorological conditions present at the onset of wildfire events during the period 2000–2025. Our findings report the climatological anomalous (climate change) influence of: temperature, wind speed, and humidity on wildfire duration. The machine learning techniques successfully validated our results, with the FC2 method based on Grammatical Evolution for constructing new features, achieving the best overall performance with mean relative error across all three regions at 1.51%. Specifically, the error rate for the Attica region was 1.90% (98.10% accuracy), for Crete 2.18% (97.82% accuracy), and for the Eastern Macedonia & Thrace region 0.46% (99.54% accuracy).

Keywords: Climate change; machine learning; neural networks

1. Introduction

This study aims to investigate the connection between climate change, specifically the factors of rising temperatures, low humidity, and high wind speed and their anomalous influence on forest fire duration. These influences will be captured through the development of a functional relationship, referred to as Final_effects. The objective is to accurately predict the predefined class of climatic effects using a 30 year period (1980 - 2010 or 1992 - 2022), in accordance with the WMO Technical Regulations on climatological normals [1], based on the prevailing climatological conditions during each fire event from 2000 - 2025. At the final stage of the study, the results will be validated using machine learning techniques [2], including: NEAT, RBF, ADAM, LBFGS, and FC2.

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Preliminary analysis of the dataset reveals that, from the starting year of 2000 to the endpoint of 2025, there has been a notable increase both in the duration of forest fires and in the anomalies values of the climatic changes across the three regions under investigation. This raises well founded concerns, cause it is well established that rising environmental temperatures extend the fire season into periods of the year when such phenomena were previously uncommon, including the winter and autumn months.

A noteworthy example of this is Canada, in 2023, experienced its warmest and driest conditions since 1980, which in turn fueled exceptionally intense wildfires that persisted for nearly five months, according to NASA[3]. Continuing this discussion, a study on the 2025 Los Angeles wildfires indicated that Climate change has increased the likelihood of the conditions that fueled these destructive fires by roughly 35 percent compared with a world that had not yet experienced industrial era warming[4]. Concerning Europe, data from the Copernicus Climate Change Service indicates that Europe experienced its hottest summer on record in 2025. Globally, the June – August 2025 period also marked the warmest summer ever observed, with temperatures rising 0.7 °C above the 1991–2020 average. The risk of wildfires is projected to rise even further as a consequence of ongoing climate change[5].

In the following, we will present relevant studies that examine the relationship between climatological phenomena and wildfire behavior. In 1988, a study examined how meteorological variables relate to the monthly area burned by wildfires during the May–August period from 1953 to 1980 across nine Canadian provinces[6]. In 1993, the following study demonstrated that fire histories that exhibit regional synchrony highlight the critical role of climate in sustaining none equilibrium dynamics [7]. In 1998, an article outputs that Climate change, as projected, poses significant challenges for forest fire management in both Canada and Russia[8]. In 2000, a study explored how climate change influences forest fire activity and discusses the subsequent implications for forest ecosystems across the United States [10]. Subsequently, in 2001, a study was published showing that, as a consequence of climate change, forests may soon experience rapid shifts in the timing, intensity, frequency, and extent of disturbances [9]. In 2006, it was again emphasized that weather and climate are the primary factors influencing fire activity, and that these factors are changing due to human induced climate change [11]. In 2011, it was reported that shifting climate conditions, marked by an average increase in minimum temperatures of approximately 1.3°C and a 15% reduction in winter precipitation, indicate an intensified risk of forest fires in the future [12]. In 2012, a study was published on Climate change and disruptions to global fire activity [13]. In 2017, a further study emphasized that fire danger is directly influenced by weather and climatic conditions [14]. In 2019, researchers developed a simulation model projecting conditions in 2060, which highlighted that climate change heightens the probability of extreme wildfire events [15]. In 2020, a study focused on reviewing a large body of research published since 2013 on climate change and forest fires, concluding that all studies consistently demonstrate a clear linkage between climate change and wildfire activity [16]. Similarly, a subsequent study published in 2022 reviewed the current understanding of how climate change affects fire weather [17]. Likewise, a 2024 study illustrates the growing vulnerability of forests to fire disturbances under climate change [18]. Finally, in 2025, an article was published investigating the loss of forested land resulting from extreme wildfire events intensified by climate change [19]. In closing the overview of studies related to our own, we now proceed with a brief summary of the main types of forest fires and their key characteristics. In this context, and given that the present research focuses exclusively on wildfires occurring within forested areas, only the categories relevant to forest land fire behavior, will be presented.

There are three principal categories of wildfires: surface fires, ground fires, and crown fires [20].

1. Surface fires are the most widespread, consuming fuels such as heather, grasses, bracken, and gorse. These fires can burn intensely, advance rapidly with extended flame lengths, and reach high levels of fire intensity. Such fuels often accumulate along roadsides and forest tracks, during the early stages before canopy closure, and following thinning operations.
2. Ground fires burn through peat and soil organic material, posing a significant threat to carbon reserves. Smouldering peat fires are particularly difficult to suppress and can reignite repeatedly.
3. Crown fires are less common and typically occur during periods of extreme heat and drought, yet they represent the most hazardous form of wildfire. They develop when surface fires spread upward through ladder fuels, such as: tall shrubs, low tree canopies, standing deadwood, and trees weakened or destabilized by wind, thereby increasing the likelihood of fire reaching and propagating through the forest canopy.

The remaining of this article is organized as follows: in section 2 the used datasets are fully described, in section 3 the experimental results from the application of various machine learning methods to the described datasets are listed and finally in section 4 a series of conclusions are presented.

2. Materials and Methods

2.1. The dataset employed

For the purposes of our study, we combined two distinct data sources.

On the one hand, to analyze the climate change conditions, we employed Historical Weather Data from Visual Crossing (<https://www.visualcrossing.com/> last accessed on 25 June 2026), using a 30 year period (1980 - 2010 or 1992 - 2022) as a baseline[1]. Beyond these baseline periods, we additionally retrieved climatological data up to the year 2025. The data compiles information from a range of authoritative sources, including national meteorological agencies, global climate datasets, and satellite observations. Access to the historical dataset required completing the registration process, after which we selected the data access interface, specified the region of interest, and defined the temporal range. Data retrieval was then carried out under the free access tier, which allows up to 1000 records per day. The procedure is generally straightforward, with the only notable constraint being the daily limit of 1000 data points.

On the other hand, for the analysis of forest / wild fires, we relied on the open data provided by the Hellenic Fire Service (https://www.fireservice.gr/el_GR/synoladedomenon last accessed on 25 June 2026). The fire data are collected and updated each year by the individual fire stations responsible for reporting incidents.

2.2. The dataset description

For this study, we obtained historical meteorological data for the period 1980 – 2025, providing a roughly 45 year record to support the effects of climate change on fire duration according. We additionally selected regions across different latitudes and longitudes, in Greece, to achieve a more comprehensive representation of the study areas. The selected regions of interest are: Attica, Eastern Macedonia and Thrace, and Crete. Before proceeding with the merging of the two data files, the experiments focused exclusively on forest fire incidents in order to maintain homogeneity in our dataset with respect to fire behavior characteristics. In other words, we chose to proceed with the following categories: (9) Burned forests, (10) Burned forest area, and (11) Burned grove (small forest) Table 2

Following this refinement, the merged climatic and fire dataset for all regions covers the period from January 1, 2000, to December 31, 2025. Specifically, Attica provides 2.100 forest fire records, Crete provides 2.893 wildfire record and finally Eastern Macedonia and Thrace provides 2.919 forest fire records.

The integration (data merge) of climate and wildfire-event datasets was performed programmatically in Python using OpenPyXL. A date-based join at daily resolution was applied per region (Attica, Crete, and Eastern Macedonia & Thrace), producing unified Excel datasets that combine meteorological predictors from Table 1 with wildfire event attributes from Table 2.

The following tables, present the primary raw features obtained from Visual Crossing and the Hellenic Fire Service.

Table 1. Definition of raw climatological variables and units of measurement

Number	Data Description	Values
1	Region	Area
2	Date	Year/month/day
3	Tempmax	°F
4	Tempmin	°F
5	Temp	°F
6	Feelslikemax	°F
7	Feelslikemin	°F
8	Feelslike	°F
9	Dew	°F
10	Humidity	%
11	Precipitation	in
12	Precipprob	%
13	Precipcover	%
14	Preciptype	Type(s)
15	Snow	in
16	Snowdepth	in
17	Windgust	mph
18	Windspeed	mph
19	Winddir	(°)
20	Sealevel	mb
21	Cloudcover	%
22	Visibility	mi
23	Solarradiation	(W/m ²)
24	Solarenergy	(MJ/m ²)
25	Uvindex	(0-10)
26	Severisk	(0-100)
27	Sunrise	time
28	Sunset	time
29	Moonphase	(0-1)
30	Conditions	general observations
31	Description	general observations day/night
32	Icon	summarized observations
33	Stations	Code for stations

Table 2. Raw wildfire incident data (Hellenic Fire Service)

Number	Data description
1	Fire department
2	Forest department
3	Location
4	Region
5	Date of fire ignition
6	Time of fire ignition
7	Extinguishment date
8	Extinguishment time
9	Burned forests
10	Burned forest area
11	Burned grove (small forest)
12	Burned Grassland
13	Burned swamps
14	Burned agricultural
15	Burned crops
16	Burned garbage dumps

Accordingly, we created three new datasets, as in Table 3 one for each study region. Each region was analyzed using its own climatological baseline, derived from the corresponding 30 year dataset. These datasets were classified into climatological normal and anomalous conditions according to their long term monthly distributions. Subsequently, we incorporated the contemporary meteorological conditions present at the time of wildfire ignition, together with the operational firefighting data recorded during suppression activities.

In the merged datasets, new columns were generated containing the normal values derived from the baseline climatological periods of previous decades. These values were then compared with the meteorological and operational firefighting conditions recorded for each event, in order to construct the final classification variable *Final_effects*. This variable quantifies, with binary values (0/1), the influence of the selected features, namely the normal versus anomalous temperature values, humidity, and wind speed, on wildfire duration.

Accordingly, the newly processed datasets for each region were structured around a specific set of features, enabling us to quantify the degree to which these characteristics influenced wildfire duration. This procedure ultimately supported the development of the *Final_effects* classification variable.

Table 3. Derived classification variables for climatic anomalies and fire duration, by region

Number	Features
1	Class_temp_anomaly (0/1)
2	Class_high_heat_anomaly (0/1)
3	Class_low_heat_anomaly (0/1)
4	Class_humidity_anomaly (0/1)
5	Class_high_humidity_anomaly (0/1)
6	Class_low_humidity_anomaly (0/1)
7	Class_windspeed_anomaly (0/1)
8	Class_high_windspeed_anomaly (0/1)
9	Class_low_windspeed_anomaly (0/1)
10	Fire_duration_class (0/1)
11	Final_effects (0/1)

2.3. Attica Data Description

For the region of Attica, the baseline period (1980–2010) indicated that the normal climatological values are those presented in the following Table 4.

Table 4. Climatological normal (baseline) values for Attica, 1980–2010

Month	Temp-max (°F)	Norm_base (Low) humidity start (%)	Norm_base (High) humidity start (%)	Norm_base (Low) windspeed start (mph)	Norm_base (High) windspeed start (mph)
January	55.52	67.50 (%)	71.10(%)	14.16 mph	15.45 mph
February	55.81				
March	60.14	56.88 (%)	65.93 (%)	14.53 mph	15.13 mph
April	67.19				
May	76.69				
June	85.68	45.88 (%)	50.86 (%)	15.40 mph	16.60 mph
July	90.81				
August	90.83				
September	83.23	54.35 (%)	69.49 (%)	13.95 mph	14.89 mph
October	73.95				
November	64.46				
December	57.69	67.50 (%)	71.10 (%)	14.16 mph	15.45 mph

In the following Table 5, the number of values is reported for each feature, distinguishing between normal and anomalous conditions.

Table 5. Distribution of normal vs. anomalous values per feature – Attica

Total data	Features	Normal (0)	Anomalous (1)
2.100	Class_temp_anomaly (0/1)	504	1.596
	Class_high_heat_anomaly (0/1)	1.063	1.037
	Class_low_heat_anomaly (0/1)	1.540	560
	Class_humidity_anomaly (0/1)	445	1.655
	Class_high_humidity_anomaly (0/1)	1.666	434
	Class_low_humidity_anomaly (0/1)	879	1.221
	Class_windspeed_anomaly (0/1)	143	1.957
	Class_high_windspeed_anomaly (0/1)	1.123	977
	Class_low_windspeed_anomaly (0/1)	1.120	980
	Fire_duration_class (0/1)	1.216	884
	Final_effects (0/1)	865	1.235

A detailed analysis of the influence of each feature will follow on Fire_duration_class (0/1), Table 6. For example: the influence of the feature **Class_high_heat_anomaly**, which contains 1.037 anomalous values, is assessed as follows. When the filter is set to (1) indicating anomalous conditions, we identify in the column **Fire_duration_class** how many corresponding values are equal to (1), and subsequently how many are equal to (0), as shown in the column (Common intersection of values). The difference between these counts represents the percentage influence of the feature on wildfire duration.

Table 6. Influence of climatic anomalies on fire duration – Attica

Features	Common intersection of values	Value division	Result expressed as a percentage (%)	Effects on fire duration (%)
Class_temp_anomaly (1) + Fire_duration_class (1)	656	656/1.596	41.10%	-17.85%
Class_temp_anomaly (1) + Fire_duration_class (0)	941	941/1.596	58.95%	
Class_high_heat_anomaly (1) + Fire_duration_class (1)	431	431/1.037	41.56%	-16.87%
Class_high_heat_anomaly (1) + Fire_duration_class (0)	606	606 / 1.037	58.43%	
Class_low_heat_anomaly (1) + Fire_duration_class (1)	225	225 / 560	40.18%	-19.64%
Class_low_heat_anomaly (1) + Fire_duration_class (0)	335	335 / 560	59.82%	
Class_humidity_anomaly (1) + Fire_duration_class (1)	731	731 / 1.655	44.17%	-11.66%
Class_humidity_anomaly (1) + Fire_duration_class (0)	924	924 / 1.655	55.83%	
Class_high_humidity_anomaly (1) + Fire_duration_class (1)	129	129 / 434	29.72%	-40.55%
Class_high_humidity_anomaly (1) + Fire_duration_class (0)	305	305 / 434	70.27%	
Class_low_humidity_anomaly (1) + Fire_duration_class (1)	602	602 / 1.221	49.30%	-1.40%
Class_low_humidity_anomaly (1) + Fire_duration_class (0)	619	619 / 1.221	50.69%	
Class_windspeed_anomaly (1) + Fire_duration_class (1)	820	820 / 1.957	41.90%	-16.19%
Class_windspeed_anomaly (1) + Fire_duration_class (0)	1.137	1.137 / 1.957	58.09%	
Class_high_windspeed_anomaly (1) + Fire_duration_class (1)	481	481 / 977	49.23%	-1,53%
Class_high_windspeed_anomaly (1) + Fire_duration_class (0)	496	496 / 977	50.76%	
Class_low_windspeed_anomaly (1) + Fire_duration_class (1)	339	339 / 980	34.59%	-30.81%
Class_low_windspeed_anomaly (1) + Fire_duration_class (0)	641	641 / 980	65.40%	

From the feature analysis, the duration of wildfires in the Attica region is influenced by -40.55% by high humidity anomalies, as in Table 4 (Class_high_humidity_anomaly). The next most influential factor is wind speed, with the feature (Class_low_windspeed_anomaly) accounting for -30.81% of the observed impact. Another feature, it is worth noting the temperature with (Class_low_heat_anomaly) at -19.64%, (Class_temp_anomaly) at -17.85%, and (Class_high_heat_anomaly) at -16.87%.

Table 8. Distribution of normal vs. anomalous values per feature – Crete

Total data	Features	Normal (0)	Anomalous (1)
2.893	Class_temp_anomaly (0/1)	865	2.028
	Class_high_heat_anomaly (0/1)	1.308	1.585
	Class_low_heat_anomaly (0/1)	2.450	443
	Class_humidity_anomaly (0/1)	396	2.497
	Class_high_humidity_anomaly (0/1)	1.322	1.571
	Class_low_humidity_anomaly (0/1)	1.967	926
	Class_windspeed_anomaly (0/1)	428	2.465
	Class_high_windspeed_anomaly (0/1)	1.468	1.425
	Class_low_windspeed_anomaly (0/1)	1.853	1.040
	Fire_duration_class (0/1)	1.568	1.325
	Final_effects (0/1)	1.068	1.825

2.4. Crete Data Description

For the region of Crete, the baseline period (1980–2010) indicated that the normal climatological values are those presented in the following Table

Table 7. Climatological normal (baseline) values for Crete, 1980–2010

Month	Temp-max (°F)	Norm_base (Low) humidity start (%)	Norm_base (High) humidity start (%)	Norm_base (Low) windspeed start (mph)	Norm_base (High) windspeed start (mph)
January	58.83	68.97 (%)	69.50 (%)	18.86 mph	20.45 mph
February	58.62				
March	61.72	62.73 (%)	66.02 (%)	15.74 mph	18.87 mph
April	67.43				
May	73.35				
June	80.25	58.77 (%)	61.72 (%)	15.92 mph	18.38 mph
July	83.34				
August	83.30				
September	79.66	64.04 (%)	68.83 (%)	15.75 mph	17.58 mph
October	74.03				
November	67.33				
December	61.66	68.97 (%)	69.50 (%)	18.86 mph	20.45 mph

In the following Table 8 the number of values is reported for each feature, distinguishing between normal and anomalous conditions.

A detailed analysis of the influence of each feature will follow, in Table 9 on Fire_duration_class (0/1).

Table 9. Influence of climatic anomalies on fire duration – Crete

Features	Common intersection of values	Value division	Result expressed as a percentage (%)	Effects on fire duration (%)
Class_temp_anomaly (1) + Fire_duration_class (1)	916	916 / 2.028	45.17%	-9.66%
Class_temp_anomaly (1) + Fire_duration_class (0)	1.112	1.112 / 2.028	54.83%	
Class_high_heat_anomaly (1) + Fire_duration_class (1)	761	761 / 1.585	48.01%	-3,97%
Class_high_heat_anomaly (1) + Fire_duration_class (0)	824	824 / 1.585	51.98%	
Class_low_heat_anomaly (1) + Fire_duration_class (1)	155	155 / 443	34.99%	-30.02%
Class_low_heat_anomaly (1) + Fire_duration_class (0)	288	288 / 443	65.01%	
Class_humidity_anomaly (1) + Fire_duration_class (1)	1.138	1.138 / 2.497	45.57%	-8.85%
Class_humidity_anomaly (1) + Fire_duration_class (0)	1.359	1.359 / 2.497	54.42%	
Class_high_humidity_anomaly (1) + Fire_duration_class (1)	722	722 / 1.571	45.96%	-8.09%
Class_high_humidity_anomaly (1) + Fire_duration_class (0)	849	849 / 1.571	54.05%	
Class_low_humidity_anomaly (1) + Fire_duration_class (1)	416	416 / 926	44.92%	-10.15%
Class_low_humidity_anomaly (1) + Fire_duration_class (0)	510	510 / 926	55.07%	
Class_windspeed_anomaly (1) + Fire_duration_class (1)	1.117	1.117 / 2.465	45.31%	-9.37%
Class_windspeed_anomaly (1) + Fire_duration_class (0)	1.348	1.348 / 2.465	54.68%	
Class_high_windspeed_anomaly (1) + Fire_duration_class (1)	712	712 / 1.425	49.96%	-0.07%
Class_high_windspeed_anomaly (1) + Fire_duration_class (0)	713	713 / 1.425	50.03%	
Class_low_windspeed_anomaly (1) + Fire_duration_class (1)	405	405 / 1.040	38.94%	-22.11%
Class_low_windspeed_anomaly (1) + Fire_duration_class (0)	635	635 / 1.040	61.05%	

From the feature analysis, the duration of wildfires in the Crete region, is influenced by (Class_low_heat_anomaly) at - 30.02% , and from (Class_low_windspeed_anomaly) at -22.11%, according to Table 9. There is also another feature it is worth noting the (Class_low_humidity_anomaly) at -10.15%.

2.5. Eastern Macedonia & Thrace Data Description

For the region of Eastern Macedonia and Thrace, the baseline period was (1992–2022) because climatological records for the region are not available prior to 1992, and the earliest data begin in 1992 rather than 1980, we adopted the 1992–2022 period as the 30 year baseline, in accordance with the WMO Technical Regulations on climatological normals. Consequently, the normal climatological values are those presented in the following Table 10.

Table 10. Climatological normal (baseline) values for Eastern Macedonia & Thrace, 1992–2022

Month	Temp max (°F)	Norm_base (Low) humidity start (%)	Norm_base (High) humidity start (%)	Norm_base (Low) windspeed start (mph)	Norm_base (High) windspeed start (mph)
January	49.04	75.45 (%)	77.98 (%)	10.51 mph	11.71 mph
February	51.28				
March	55.93	69.15 (%)	74.02 (%)	11.60 mph	12.71 mph
April	63.61				
May	73.50	62.18 (%)	67.14 (%)	10.80 mph	11.11 mph
June	81.28				
July	86.04	66.59 (%)	77.56 (%)	10.57 mph	11.25 mph
August	86.90				
September	78.76	75.45 (%)	77.98 (%)	10.51 mph	11.71 mph
October	69.05				
November	59.74				
December	51.74				

In the following Table 11, the number of values is reported for each feature, distinguishing between normal and anomalous conditions.

Table 11. Distribution of normal vs. anomalous values per feature – Eastern Macedonia & Thrace

Total data	Features	Normal (0)	Anomalous (1)
2.919	Class_temp_anomaly (0/1)	677	2.242
	Class_high_heat_anomaly (0/1)	1.289	1.630
	Class_low_heat_anomaly (0/1)	2.307	612
	Class_humidity_anomaly (0/1)	684	2.235
	Class_high_humidity_anomaly (0/1)	2.267	652
	Class_low_humidity_anomaly (0/1)	1.336	1.583
	Class_windspeed_anomaly (0/1)	38	2.881
	Class_high_windspeed_anomaly (0/1)	1.514	1.405
	Class_low_windspeed_anomaly (0/1)	1.443	1.476
	Fire_duration_class (0/1)	1.933	986
	Final_effects (0/1)	1.124	1.795

A detailed analysis of the influence of each feature will follow, in Table 12 on Fire_duration_class (0/1).

Table 12. Influence of climatic anomalies on fire duration – Eastern Macedonia & Thrace

Features	Common intersection of values	Values division	Result expressed as a percentage (%)	Effects on fire duration (%)
Class_temp_anomaly (1) + Fire_duration_class (1)	759	759 / 2.242	33.85%	-32.29%
Class_temp_anomaly (1) + Fire_duration_class (0)	1.483	1.483 / 2.242	66.14%	
Class_high_heat_anomaly (1) + Fire_duration_class (1)	594	594 / 1.630	36.44%	-27.11%
Class_high_heat_anomaly (1) + Fire_duration_class (0)	1.036	1.036 / 1.630	63.55%	
Class_low_heat_anomaly (1) + Fire_duration_class (1)	165	165 / 612	26.96%	-46.07%
Class_low_heat_anomaly (1) + Fire_duration_class (0)	447	447 / 612	73.03%	
Class_humidity_anomaly (1) + Fire_duration_class (1)	759	759 / 2.235	33.96%	-32.08%
Class_humidity_anomaly (1) + Fire_duration_class (0)	1.476	1.476 / 2.235	66.04%	
Class_high_humidity_anomaly (1) + Fire_duration_class (1)	168	168 / 652	25.77%	-48.46%
Class_high_humidity_anomaly (1) + Fire_duration_class (0)	484	484 / 652	74.23%	
Class_low_humidity_anomaly (1) + Fire_duration_class (1)	591	591 / 1.583	37.33%	-25.33%
Class_low_humidity_anomaly (1) + Fire_duration_class (0)	992	992 / 1.583	62.66%	
Class_windspeed_anomaly (1) + Fire_duration_class (1)	974	974 / 2.881	33.81%	-32.38%
Class_windspeed_anomaly (1) + Fire_duration_class (0)	1.907	1.907 / 2.881	66.19%	
Class_high_windspeed_anomaly (1) + Fire_duration_class (1)	502	502 / 1.405	35.73%	-28.54%
Class_high_windspeed_anomaly (1) + Fire_duration_class (0)	903	903 / 1.405	64.27%	
Class_low_windspeed_anomaly (1) + Fire_duration_class (1)	472	472 / 1.476	31.98%	-36.04%
Class_low_windspeed_anomaly (1) + Fire_duration_class (0)	1.004	1.004 / 1.476	68.02%	

From the feature analysis, the duration of wildfires in the Eastern Macedonia & Thrace region is most strongly influenced by humidity, with (Class_high_humidity_anomaly) accounting for -48.46%, according to Table 10. Temperature also plays a substantial role, with (Class_low_heat_anomaly) at -46.07%, followed by wind speed, with (Class_low_windspeed_anomaly) at -36.04%. Other features exhibit comparable secondary effects, including (Class_windspeed_anomaly) at -32.38%, (Class_temp_anomaly) at -32.29%, and (Class_humidity_anomaly) at -32.08%.

3. Results

For the purposes of our experiments, the validation of our findings was performed using the available optimization package Optimus [21] which can be downloaded from <https://github.com/itsoulos/GlobalOptimus> (accessed on 15 July 2026). With Grammatical Evolution [22], new features for FC2 were constructed [23] through an evolutionary process, and we utilized the freely available programming tool QFc [24], to implement this procedure which can be downloaded from <https://github.com/itsoulos/QFc> (accessed on 15 July 2026). In addition, the Weka [25] environment was used for the experiments involving the Radial Basis Function (RBF) network.

3.1. The Feature Construction Method

The proposed feature construction method, aims at the automatic generation of new features from the original ones through evolutionary processes, using the Grammatical Evolution procedure [26]. Specifically, the constructed features (FC) represent nonlinear combinations of the original ones and are subsequently used for training a machine learning model. The model's performance on the training data serves as the fitness function for the Grammatical Evolution process, providing feedback that guides the evolutionary search toward generating more effective features. This method has been applied across many different domains, including disease prediction[27], wildfire duration estimation[28], Parkinson's symptom detection[29], object recognition[30], image quality enhancement[31], and several others. The feature construction (FC) process involves two main steps.

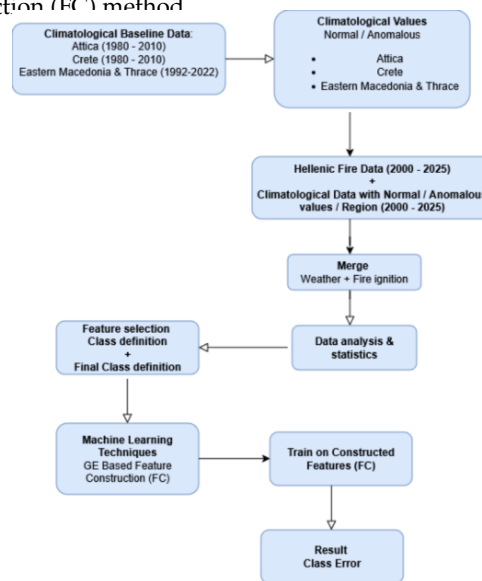
1. The feature construction phase, in which candidate feature sets are explored using Grammatical Evolution as the primary mechanism. This evolutionary procedure (selection, crossover, mutation) is iterated over a number of generations, aiming to identify the genotype that yields the highest performing features.
2. The feature evaluation phase, during which the best genotype is used to generate the new set of (N_f) features. This step ensures a rigorous assessment based on their performance.

Below, the Figure 1 illustrates the workflow carried out in this study, aimed at identifying the influence of the climatic characteristics that, in practice, affect wildfire duration, as well as validating our results through the feature construction (FC) technique, which ultimately produced the best performance. Specifically, we first constructed the climatological baseline for each region, identifying the normal and anomalous values. Subsequently, we processed the data provided by the Hellenic Fire Service, focusing exclusively on forest area wildfires, their duration, and the day of occurrence. Using the ignition day as the reference point, we merged the corresponding meteorological records from the same calendar dates. Based on the baseline analysis, we then identified the climatic attributes required to build the dataset, defined the binary classes (0/1), and established the final output (target) class.

Afterwards, we processed the resulting datasets using machine learning techniques. Among these methods, the feature construction (FC) approach stood out: feature construction is performed by the proposed GE based FC, which searches for nonlinear expressions of the original variables and produces a compact set of constructed features (N_f) tailored to the classification task. These features are then trained and evaluated using classification error (%) as the primary metric.

Each experiment was executed 30 times, using different seeds for the random number generator in every run. In addition, the ten fold cross - validation procedure was applied to ensure robust validation of the experimental results. The proposed feature construction method was applied independently to each fold, and the average classification errors on the corresponding test sets were computed.

Figure 1. Workflow of the proposed methodology, from climatological baseline construction and merging with wildfire records, through feature and class definition, to model training and evaluation using the feature construction (FC) method



To evaluate the predictive performance of the proposed feature construction method against widely used machine learning approaches, five models were trained and tested on the climatological and wildfire duration records of the three study regions: a genetic-neuroevolution network (NEAT) [32], a radial basis function network (RBF) [33,34], a feedforward neural network trained with the Adam optimizer [35,36], a feedforward neural network trained with the L-BFGS algorithm [37], and the proposed feature construction method based on grammatical evolution (FC2). Table 13 reports the mean relative error obtained by each method for Attica, Crete, and Eastern Macedonia & Thrace, together with the average performance across all three regions.

Table 13. Experimental results on the used datasets using a series of machine learning methods.

DATASET	NEAT	RBF	ADAM	LBFGS	FC2
ATTICA	15.35%	4.81%	3.83%	2.98%	1.90%
CRETE	16.36%	1.79%	2.25%	3.70%	2.18%
MACEDONIA	14.24%	5.94%	7.56%	0.51%	0.46%
AVERAGE	15.32%	4.18%	4.55%	2.40%	1.51%

The differences among the five methods, summarized numerically in Table 13, are visualized in Figure 2 to facilitate a direct region-by-region comparison. The chart highlights that FC2 consistently ranks among the best-performing methods in Attica and Eastern Macedonia & Thrace, whereas in Crete the RBF network marginally outperforms it. NEAT, in contrast, exhibits substantially higher error across all three regions, underlining its limited suitability for this prediction task.

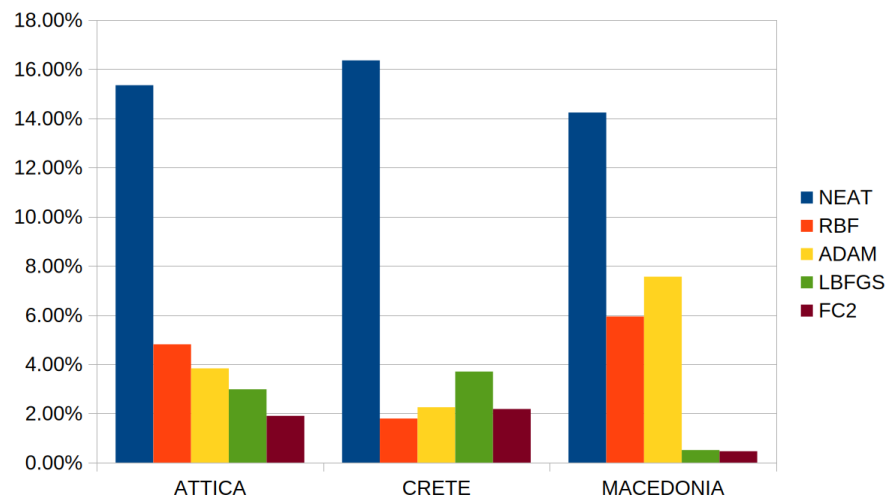


Figure 2. Graphical representation of the experimental results obtained by the application of various machine learning methods.

Beyond the comparison against alternative methods, we further examined how the number of features constructed by the grammatical evolution process affects model accuracy. Figure 3 illustrates the average error obtained when one, two, or three constructed features ($F = 1, F = 2, F = 3$) are used as inputs. Across all three regions, the error decreases as additional constructed features are introduced, with the most pronounced improvement observed in Eastern Macedonia & Thrace, indicating that extracting more than a single derived feature contributes measurably to the model's predictive capacity.

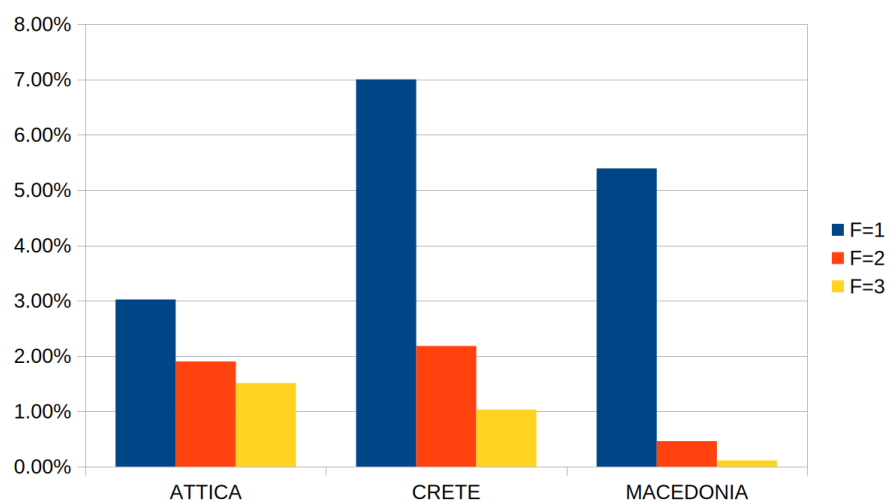


Figure 3. Graphical representation of the average classification obtained by the feature construction method using a variety of values for the number of constructed features ($F = 1, F = 2, F = 3$).

The results of the statistical analysis are presented graphically in Figure 4, where the mean error and standard deviation of each method are shown alongside the corresponding significance level relative to FC2. Although the Friedman test did not reach the conventional 0.05 significance threshold (chi-squared = 8.533, $p = 0.074$), the Nemenyi critical difference of 3.522 confirms that the average rank of FC2 differs significantly from that of NEAT, consistent with the paired t-test result ($p = 0.0002$). The remaining comparisons, against LBFGS, RBF, and ADAM, do not reach statistical significance, which is likely attributable to the small number of regions ($n = 3$) available for the paired tests.

Model	Mean Error	SD Error	Min Error	Max Error	Avg Rank	Mean Diff vs FC2	Wilcoxon p	Paired t_p	Significance
FC2	1.51	0.92	0.46	2.18	1.33	0			Reference
LBFGS	2.4	1.67	0.51	3.7	2.67	-0.88	0.1814	0.1798	n.s.
RBF	4.18	2.15	1.79	5.94	2.67	-2.67	0.4227	0.2571	n.s.
ADAM	4.55	2.73	2.25	7.56	3.33	-3.03	0.1814	0.286	n.s.
NEAT	15.32	1.06	14.24	16.36	5	-13.8	0.1814	0.0002	***

Table 14. Statistical comparison of the proposed FC2 method against NEAT, RBF, ADAM, and LBFGS.

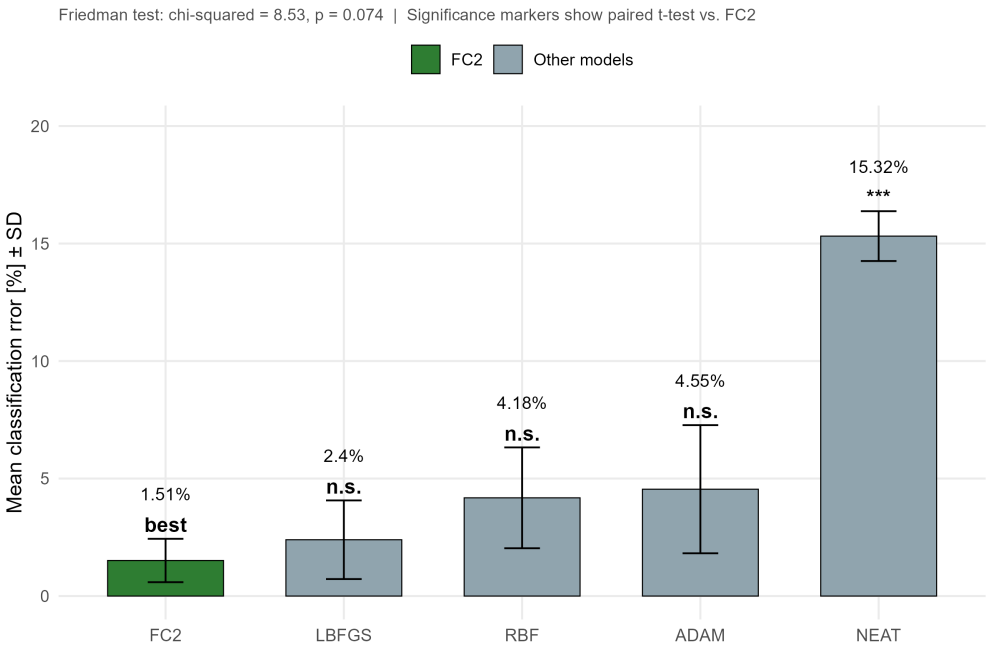


Figure 4. Mean prediction error per method with standard deviation and statistical significance relative to FC2.

To determine whether the observed differences in predictive accuracy are statistically meaningful rather than attributable to chance, a non-parametric Friedman test was applied across the three regions, followed by pairwise comparisons of FC2 against each of the remaining four methods using the paired Wilcoxon signed-rank test and the paired t-test. Table 14 summarizes the descriptive statistics, average ranks, and significance outcomes of these comparisons, using FC2 as the reference method.

Friedman test: chi-squared = 8.533, df = 4, p-value = 0.0739
Nemenyi critical difference (alpha = 0.05): 3.522
Significance vs. FC2 (paired t-test): *** p<0.001, ** p<0.01, * p<0.05, n.s. = not significant

4. Conclusions

This study developed and validated the climatological influences associated with climate change for three climatically distinct regions of Greece, namely Attica, Eastern Macedonia & Thrace, and Crete, using forty five years (1980–2025) of combined climatological and operational fire incident data. Specifically, thirty years of climatological data were used as the baseline, 1980–2010 for Attica and Crete, and 1992–2022 for Eastern Macedonia & Thrace, due to the absence of earlier historical records. Based on these data, classes were generated to define the climatological normal values for each season, which were then compared with the conditions observed during the 2000–2025 fire events. This comparison

allowed us to identify the anomalous climatological values and assess their influence on the duration of each forest fire.

Finally, the Final_effects class was generated from the combined influence of temperature, humidity, and wind speed features, and its reliability was validated using five machine learning techniques across all three datasets. The machine learning techniques that validated our results are: the NEAT method, the Radial Basis Function (RBF), the LBFGS optimizer, the ADAM optimizer, and the Feature Construction (FC2). The machine learning techniques successfully validated our results, with the FC2 method based on Grammatical Evolution for constructing new features, achieving the best overall performance with mean relative error across all three regions at 1.51%. Specifically, the error rate for the Attica region was 1.90% (98.10% accuracy), for Crete 2.18% (97.82% accuracy), and for the Eastern Macedonia & Thrace region 0.46% (99.54% accuracy). It is worth noting that only for the region of Crete did the RBF achieve a slightly better performance, with an error rate of 1.79%.

In addition, the statistical validation reinforces that the difference between FC2 and NEAT is unlikely to be attributable to chance. The overall Friedman test did not reach the conventional 0.05 significance threshold (chi-squared = 8.533, $p = 0.074$), and the Nemenyi critical difference of 3.522 confirms that the average rank of FC2 differs significantly only from that of NEAT, while the differences from RBF, LBFGS, and ADAM are not statistically significant. This outcome is consistent with the paired t-test, which likewise shows a statistically significant difference from NEAT ($p = 0.0002$), but not from the remaining three methods.

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